PROITBRIDGE
Strive For Better Future

Core NLP Program

(Project-Based Learning)

06 CHAPTERS | 14 PROJECTS | 16 TOOLS

10 DAYS

Project-Based Learning (8 Industries We Tap)

1. **E-Commerce & Retail** — Feedback analysis · recommendations
2. **Media & News** — News classification · fake news detection
3. **Recruitment & HR** — Resume screening · skill extraction
4. **Customer Support** — Spam filtering · issue & intent detection
5. **Healthcare** — Diseases · medicines · symptoms
6. **Banking & Insurance** — Transaction info · claim details
7. **Legal & Compliance** — Contract parties · dates · clauses
8. **Search & Knowledge** — Semantic search · summarization

CHAPTER 1

NLP Foundations, Preprocessing & Vectorization

From raw text to machine-ready numerical features

TOOLS

Python · Pandas · NumPy · Regex · NLTK · spaCy · Scikit-Learn

TOPICS

- Introduction to NLP
- Text Cleaning & Lower Casing
- Tokenization
- Preprocessing Pipeline
- Bag of Words
- N-Grams
- NLP Pipeline
- Punctuation & Stop Words
- Stemming & Lemmatization
- Corpus & Vocabulary
- Count Vectorizer
- TF · IDF · TF-IDF

PROJECTS

1. **Customer Feedback Analysis System**
2. **Email Spam Detection System**
3. **News Classification System**

→ Prepare raw text for ML models → Convert text into numerical representations

CHAPTER 2

Word Embeddings & Semantic Search

From sparse vectors to meaning — how modern search works

TOOLS

Scikit-Learn · Gensim · Word2Vec · FastText

TOPICS

- Limitations of TF-IDF
- Word Embeddings
- CBOW & Skip-Gram
- Cosine Similarity
- Document Similarity
- Dense Vectors
- Word2Vec
- FastText
- Semantic Similarity
- Vector Representation

PROJECTS

4. **Document Similarity Engine**
5. **Semantic Search System**
6. **Product Recommendation Similarity System**

→ Measure similarity between documents → Understand semantic meaning & context

CHAPTER 3

Deep Learning for NLP

How machines learn sequential language patterns

TOOLS

TensorFlow · Keras

TOPICS

- Sequence Data
- RNN
- GRU
- Sequence Prediction
- Embedding Layers
- LSTM
- Vanishing Gradients
- Text Generation

PROJECTS

7. **Sentiment Classification System**
8. **Next Word Prediction System**

→ Learn sequential language patterns → Build prediction-based NLP systems

CHAPTER 4

Transformers & BERT

The architecture behind ChatGPT, Gemini and modern LLMs

TOOLS

Hugging Face · Transformers · PyTorch

TOPICS

- Attention Mechanism
- Multi Head Attention
- Encoder · Decoder
- BERT Architecture
- Self Attention
- Positional Encoding
- Transformer Architecture
- Transfer Learning

PROJECTS

9. **Language Translation System**
10. **Text Summarization System**

→ Understand the foundation behind modern LLMs → See how transformers process language

CHAPTER 5

NER & Entity Extraction

Teaching machines to find entities inside documents

TOOLS

Hugging Face · Transformers · Datasets · PyTorch

TOPICS

- Named Entity Recognition
- Information Extraction
- Tokenizers
- Entity Extraction
- Fine Tuning for NER
- Entity-Level Evaluation

PROJECTS

11. **Named Entity Recognition System**
12. **Resume Screening System**

→ Extract entities from text → The base of RAG & agent memory

CHAPTER 6

Other Industry NLP Applications

Production-grade NLP systems and workflows

TOOLS

Hugging Face · Transformers · Datasets · PyTorch

TOPICS

- Question Answering
- Inference Pipelines
- Fine Tuning
- Text Classification
- Model Saving
- Industry NLP Workflows

PROJECTS

13. **Question Answering System**
14. **Fake News Detection System**

→ Ship end-to-end solutions → Ready for LLMs & RAG

Core NLP foundations are a must before Agentic AI.

CORE NLP → LLMs → RAG → AGENTIC AI